BSCIT 375: Knowledge Discovery and Data Science Assignment-I

Submit a handwritten solution for each of the following questions.

Show all steps in calculations where applicable.

Module 1: Introduction to KDD and Data Science

Q1 Distinguish between Knowledge Discovery in Databases (KDD) and Data Mining. Provide a real-world example of each.

Q2 List and briefly describe the six phases of the CRISP-DM methodology.

Q3 What are the main differences between CRISP-DM and SEMMA? Create a comparison table.

Q4 Define the following data types with examples: nominal, ordinal, continuous, discrete.

Q5 Why is data preprocessing considered the most critical step in the knowledge discovery process?

Q6 Explain the concept of a target variable. How does it influence the choice between supervised and unsupervised learning?

Q7 Calculate the mean, median, and mode for the following dataset: {12, 15, 12, 18, 22, 15, 19, 15}.

Q8 Compute the range, variance, and standard deviation for the sample: {5, 10, 15, 20, 25}.

Q9 What is the Interquartile Range (IQR)? How is it used to detect outliers?

Q10 Using the empirical rule (68-95-99.7), what percentage of data falls within 2 standard deviations of the mean in a normal distribution?

Q11 A dataset has mean = 100 and standard deviation = 15. Find the Z-score for a value of 130. Is it an outlier under the $|Z| > 3$ criterion?

Q12 Explain the difference between Min-Max normalization and Z-score standardization. When would you prefer one over the other?

Q13 Apply Min-Max normalization to the values [50, 100, 150, 200] to scale them to the range [0,1].

Q14 A feature has values [10, 20, 30, 40, 1000]. Why would Min-Max scaling be problematic? How does standardization handle this better?

Module 2: Data Preprocessing and EDA

Q15 Define MCAR, MAR, and MNAR with examples. How does the handling of missing data differ for each?

Q16 List three methods for imputing missing numerical data. Discuss one advantage and one disadvantage of each.

Q17 Using the Z-score method, identify outliers in the set: $\{2, 3, 5, 7, 9, 12, 45\}$. Assume mean = 11.86, sd 14.5.

Q18 Using the IQR method, find outliers in: $\{10, 12, 14, 18, 22, 24, 30, 35, 100\}$. Show Q1, Q3, IQR, and fences.

Q19 Calculate the Pearson correlation coefficient for X = [1,2,3,4,5] and Y = [2,4,6,8,10]. Interpret the result.

Q20 What does a chi-square test for independence tell us? Write the formula for expected frequency.

Q21 Given the contingency table below, compute the chi-square statistic and test independence at =0.05 (critical value 3.841).

	Pass	Fail
Male	30	20
Female	25	25

Q22 Compare one-hot encoding and label encoding. When is each appropriate?

Q23 What is the dummy variable trap? How can it be avoided?

Q24 Explain Principal Component Analysis (PCA) in simple terms. How does it reduce dimensionality?

Q25 Describe three methods of numerosity reduction.

Q26 Apply equal-width binning with 3 bins to the data: [5, 7, 12, 15, 18, 22, 25, 30, 35, 40].

Q27 Apply equal-frequency binning with 3 bins to the same data set above.

Q28 What is a concept hierarchy? Give an example for a date attribute (e.g., day $\rightarrow$ month $\rightarrow$ year).

Module 3: Basic Probability and Bayes

Q29 State the three axioms of probability.

Q30 A fair die is rolled. What is the probability of getting an even number or a number greater than 4?

Q31 How many different 4-digit PINs can be formed from digits 0-9 if repetition is allowed? Not allowed?

Q32 A committee of 4 is to be chosen from 8 men and 6 women. How many ways if it must have at least 2 women?

Q33 In a lottery, you pick 5 numbers from 1 to 50. What is the probability of winning the jackpot?

Q34 Two cards are drawn without replacement from a standard deck. Find the probability both are hearts.

Q35 Define conditional probability. If P(A)=0.6, P(B)=0.4, P(AB)=0.2, find P(A—B) and P(B—A).

Q36 A test for a disease is 99

Q37 In a certain city, 30

Q38 A bag contains 3 red and 2 blue marbles. Two marbles are drawn with replacement. Find P(both red). Without replacement.

Module 4: Statistical Inference (Univariate and Multivariate)

Q40 A sample of 50 students has a mean test score of 75 with =10. Construct a 95

Q41 A sample of 20 items has mean 50 and sample standard deviation 5. Construct a 90

Q42 Explain the difference between a Z-interval and a t-interval.

Q43 In a survey of 200 people, 120 support a new policy. Find a 95

Q44 A company claims that less than 10

Q45 Two groups are given different training. Group A (n=30, mean=85, sd=4); Group B (n=35, mean=82, sd=5). Test if the means differ significantly (=0.05, use Welch's t).

Q46 Compare proportions: Sample1: 40/200, Sample2: 60/250. Test if p1 ¡ p2 at =0.05.

Q47 What are the assumptions of a chi-square test for independence? Provide a 3x2 contingency table example.

Q48 For the table below, compute expected frequencies and chi-square statistic.

	Category A	Category B	Category C
Group1	20	30	10
Group2	30	20	20

Q49 A die is rolled 60 times with results: 1:8, 2:10, 3:12, 4:9, 5:11, 6:10. Test if the die is fair (goodness-of-fit).

Q50 Explain one-way ANOVA. Write the formulas for SSB and SSW.

Q51 Given three groups with data: G1: [5,6,7], G2: [8,9,10], G3: [11,12,13]. Compute ANOVA F-statistic.

Module 5: Regression Analysis

Q52 What is the difference between correlation and regression?

Q53 Write the simple linear regression model equation. Define each term.

Q54 Derive the normal equations for the least squares estimates of slope and intercept.

Q55 Given X = [1,2,3,4,5] and Y = [2,4,5,4,5], calculate the regression line $\hat{Y}$ = a + bX.

Q56 Interpret the slope and intercept from the regression line in the previous question.

Q57 Compute the coefficient of determination (R^2) for the model in Q55.

Q58 Calculate the standard error of the estimate (s) for the same model.

Q59 Test whether the slope is significantly different from zero (=0.05) using t-test.

Q60 For the multiple regression model Y = + X + X + , explain the meaning of each coefficient.

Q61 What is multicollinearity? How is it detected using VIF?

Q62 Explain adjusted R^2. Why is it preferred over R^2 when adding predictors?

Q63 A house price model: Price = 50 + 0.2*Sqft + 10*Bedrooms. Predict price for 2000 sqft and 3 bedrooms.

Module 6: Supervised Learning – KNN, Bayes, Decision Trees, Random Forest

Q64 Compare supervised and unsupervised learning with one example of each.

Q65 What is the K-Nearest Neighbors algorithm? Explain the role of K and distance metric.

Q66 Given points A(1,2), B(2,3), C(3,3), D(6,7) and new point P(2,2). Using K=3 and Euclidean distance, classify P if classes are A,B,C as 'X' and D as 'Y'.

Q67 In KNN regression, how is the prediction made?

Q68 What are the advantages and disadvantages of KNN?

Q69 State the Naive Bayes assumption. Why is it called "naive"?

Q70 Using Laplace smoothing (=1), compute P(Sunny—Yes) from:

Outlook	Yes	No
Sunny	2	3
Overcast	4	1
Rainy	1	2

Q71 What is the Bayes decision rule? How does it minimize misclassification?

Q72 Define entropy and information gain. Write the entropy formula.

Q73 A dataset has 9 Yes and 5 No. Compute the entropy.

Q74 For a binary split: Subset1 (3 Yes, 1 No), Subset2 (6 Yes, 4 No). Compute information gain if parent entropy = 0.94.

Q75 Calculate Gini impurity for a node with 4 Yes and 6 No.

Q76 For the split in Q74, compute weighted Gini impurity.

Q77 Explain how CART handles continuous features.

Q78 What is pruning in decision trees? Differentiate pre-pruning and post-pruning.

Q79 Explain bagging. How does Random Forest improve upon a single decision tree?

Q80 What is out-of-bag (OOB) error? Why is it useful?

Q81 List two methods for measuring variable importance in Random Forest.

Q82 Compare a single decision tree and Random Forest in terms of interpretability and accuracy.

Module 7: Unsupervised Learning – K-Means and Association Rules

Q84 Describe the K-Means clustering algorithm step by step.

Q85 Given points: (1,1), (1,2), (2,1), (5,5), (5,6), (6,5). Run one iteration of K-Means with K=2 and initial centroids (1,1) and (5,5). Show assignments and new centroids.

Q86 What is the elbow method for choosing K in K-Means?

Q87 Define silhouette score. What does a value close to 1 indicate?

Q88 Define support, confidence, and lift for an association rule X Y.

Q89 Given support(Milk)=0.6, support(Bread)=0.5, support(Milk,Bread)=0.3. Compute confidence and lift for Milk Bread. Interpret lift.

Q90 What is the Apriori property? Why is it important for pruning?

Q91 From the transaction data below, find all frequent itemsets with $\min_s up = 0.6(3outof5transaction$

T1: A,B,C; T2: A,B; T3: A,C; T4: B,C; T5: A,B,C

Q92 Generate all strong association rules from the frequent itemset A,B,C with $\min_c onf = 0.7$.

Q93 What is the difference between supervised and unsupervised learning in the context of association rules?

Q94 List three measures (other than support/confidence) to evaluate rule usefulness.

Q95 Why can association rules be misleading? Provide an example of a spurious correlation.

Module 8: Model Evaluation and Performance Metrics

Q96 Draw a confusion matrix for binary classification and label TP, TN, FP, FN.

Q97 From the confusion matrix: TP=80, FN=20, FP=10, TN=90. Compute accuracy, precision, recall, and F1-score.

Q98 Why is accuracy not a good metric for imbalanced data? Give an example.

Q99 What is the ROC curve? Define TPR and FPR.

Q100 Compute the AUC from the following ROC points: (0,0), (0.2,0.7), (0.5,0.9), (1,1). Use trapezoidal rule.

Q101 What is the bias-variance trade-off? Explain with a sketch.

Q102 Describe cross-validation. Why is stratified k-fold preferred for classification?

Q103 What is overfitting? How can learning curves help detect it?

Q104 What is the purpose of a baseline model? Give two examples of simple baselines.

Q105 For an imbalanced dataset (90

Q106 A classifier has precision=0.8 and recall=0.6. Compute F1-score.

Q107 Write the formula for the cross-validation estimate of prediction error.

Module 9: Mathematics for Neural Networks

Q108 Define vector addition and scalar multiplication with examples.

Q109 Compute the dot product of u = [1,2,3] and v = [4,5,6].

Q110 Find the Euclidean norm of vector w = [3, -4, 12].

Q111 Given vectors a = [1,0] and b = [0,1], what is the angle between them?

Q112 Multiply matrices: A = [[1,2],[3,4]] and B = [[5,6],[7,8]].

Q113 Explain the geometric meaning of the dot product.

Q114 What is the Frobenius norm of a matrix? Give an example.

Q115 Define the derivative of a function. State the power rule.

Q116 Compute f/x and f/y for f(x,y) = $3x^2y$ + sin(xy).

Q117 Write the chain rule for a composite function h(x)=f(g(x)).

Q118 Compute the derivative of the sigmoid function (z) = 1/(1+e).

Q119 Why is ReLU preferred over sigmoid in deep hidden layers?

Q120 Write the formula for mean squared error (MSE) and binary cross-entropy loss.

Q121 Explain gradient descent update rule: $w_{n}ew = w_{o}ld - L/w$.

Module 10: AWS Data Pipeline and Cloud Computing

Q122 What is a data pipeline? List its typical stages.

Q123 Explain the Medallion Architecture (Bronze, Silver, Gold layers) in AWS.

Q124 What is Amazon S3? How is it used as a data lake?

Q125 Compare Amazon Kinesis Data Streams and Amazon SQS. When would you use each?

Q126 What is AWS Glue? Describe its main components (Data Catalog, Crawlers, ETL jobs).

Q127 Why is Apache Spark preferred over Pandas for large-scale data processing?

Q128 What is Amazon Athena? How does it enable serverless querying on S3?

Q129 What is the difference between Athena and Redshift?

Q130 Describe how Amazon SageMaker supports the end-to-end ML lifecycle.

Q131 In the context of SageMaker, what is a feature store?

Q132 Write a simple PySpark transformation that reads from a Glue catalog, filters records where 'age' ¿ 30, and writes to S3 in Parquet format.

Q133 How can you optimize Athena query performance and reduce costs?

Q134 List three best practices for designing data pipelines in the cloud.

Q135 How does AWS Glue use the concept of "serverless" to benefit data scientists?

Mixed Numerical Practice

Q136 Compute the Euclidean distance between (3,4) and (6,8).

Q137 Normalize the vector [100, 200, 300] using min-max to [0,1].

Q138 Calculate the median of: 5, 9, 12, 18, 22, 25, 30.

Q139 Find the IQR of: 4, 8, 15, 16, 23, 42.

Q140 A Z-score of -2.5 corresponds to what percentile in a standard normal distribution?

Q141 If P(A)=0.3, P(B)=0.5, P(AB)=0.6, find P(AB). Are A and B independent?

Q142 From the confusion matrix: TP=45, FP=15, FN=5, TN=35. Compute specificity (TNR).

Q143 For a rule X→Y, support(X)=0.4, support(Y)=0.5, support(XY)=0.2. Compute lift. Is the rule useful?

Q144 A regression model has SST=500 and SSE=100. Compute R^2.

Q145 In K-Means, after one iteration, centroids are (2,3) and (8,9). Reassign point (5,6) to the nearest centroid.